

Literature Answer to Question 3 of arXiv:1509.01499

fa_banach_001, agent_lane_12

2026-06-20

Status

This is a lightweight `literature_already_answered` packet. It records that Question 3 in the final open-question section of Flores, Hernandez, and Tradacete, *Disjointly homogeneous Banach lattices and applications*, arXiv:1509.01499, was answered by a separate later paper.

Source Question

The source paper asks, in Section “Some open questions”:

Is there a reflexive p -DH r.i. space on $[0, 1]$ whose dual is not DH?

Here p -DH means p -disjointly homogeneous and r.i. means rearrangement invariant.

Supporting Answer

The supporting paper is Sergey V. Astashkin, *Duality problem for disjointly homogeneous rearrangement invariant spaces*, arXiv:1805.00691.

Its introduction explicitly identifies the same problem, saying that it solves the duality problem and referring to Question 3 in the survey. The decisive result is Theorem 2:

For every $1 < p < \infty$ there exists a p -DH reflexive r.i. space X_p on $[0, 1]$ such that X_p^* is not DH.

This gives a positive answer to the source question.

Scope

Only Question 3 from the source paper’s final list is recorded as answered here. The source paper’s other final questions, including the complemented disjoint sequence question, the separable DH versus DC question, the symmetric atomic lattice question, and the Kato-property question, are outside this packet.

References

1. J. Flores, F. L. Hernandez, and P. Tradacete, *Disjointly homogeneous Banach lattices and applications*, arXiv:1509.01499, 2015.
2. S. V. Astashkin, *Duality problem for disjointly homogeneous rearrangement invariant spaces*, arXiv:1805.00691, 2018.